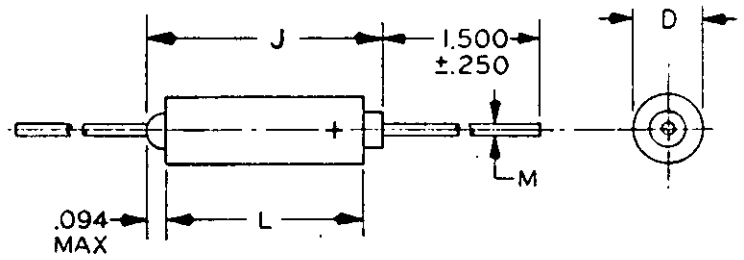
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), HIGH FREQUENCY, ESTABLISHED RELIABILITY, STYLE CSR21

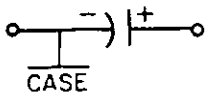
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification sheet and the latest issue of MIL-C-39003.

FAILURE RATE LEVELS "M", "P" AND "R" ARE INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1984.



CIRCUIT DIAGRAM



Case size	Dimensions			
	L ±.031	D +.016 -.015	M ±.002	J max
C	.686	.289	.025	.822
D	.786	.351	.025	.922

INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.094	2.39	.686	17.42
.015	.38	.250	6.35	.786	19.96
.016	.41	.289	7.34	.822	20.88
.025	.64	.351	8.92	.922	23.42
.031	.79			1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The case insulation shall extend .015 (.38 mm) minimum beyond each end. However, when a shrink-fitted insulation is used, it shall lap over the ends of the capacitor body.
4. Lead length may be a minimum of 1-inch long for use in tape and reel automatic insertion equipment, when specified.
5. Insulation thickness is included in dimension D.

FIGURE 1. Dimensions and configurations.

Ⓑ denotes changes

FSC 5910

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular, insulated.

Material - Metal (alloy unspecified).

Type of seal - Hermetic.

Leads: Type N of MIL-STD-1276, tin-lead plating, electrodeposited or hot-solder dipped.

DC rated voltage: See table I. Above 85°C voltage derating is required (see MIL-C-39003).

DC leakage (DCL): See table I.

Capacitance (Cap.) (nom): See table I.

Cap. tolerance: 5, 10, and 20 percent, see table I.

Dissipation factor (DF at 1 kHz): See table I.

Equivalent series resistance (ESR at 100 kHz): See table I.

Operating temperature: -55° to +125°C.

Failure rate level: See table I.

Stability at low and high temperatures: In accordance with MIL-C-39003.

Step 1 (+25°C):

DCL - See table I.

Cap. - Within tolerance specified in table I.

DF - See table I.

Step 2 (-55°C):

Cap. - Shall not change more than ±10 percent of step 1 measured value.

DF - 2 times step 1 measured value or table I whichever is greater.

Step 3 (+25°C):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of step 1 measured value.

DF - See table I.

Step 4 (+85°C):

DCL - See table I.

Cap. - Shall not change more than ±8 percent of step 1 measured value.

DF - 1.5 times step 1 measured value or table I whichever is greater.

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Step 5 (+125°C):

DCL - See table I.

Cap. - Shall not change more than ±12 percent of step 1 measured value.

DF - 1.5 times step 1 measured value or table I whichever is greater.

Step 6 (+25°C):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of step 1 measured value.

DF - See table I.

Surge voltage: In accordance with MIL-C-39003.

DCL - See table I.

Cap. - Shall not change more than ±2 percent of initial measured value.

DF - See table I.

Surge current 1/: Capacitors shall be subjected to 3 consecutive cycles at the following conditions:

Temperature - 25°C ±5°C.

Applied voltage - Rated dc voltage ±2% from a power source having an energy storage bank of 50,000 µF (minimum) across the output terminals.

Charge cycle - 4 ±1 seconds.

Discharge cycle - 4 ±1 seconds to a voltage below 1 percent of rated voltage.

The total dc resistance (excluding the capacitor) including the wiring, fixturing, and output impedance of the regulated power supply to each test position during the charging cycle shall be 1 ±0.2 ohms.

Upon completion of the test, the capacitors shall meet the following requirements:

DCL - See table I.

Cap. - Shall meet initial measured value.

DF - See table I.

Life: Method 108 of MIL-STD-202.

240, 1,000, and 2,000 hours:

At 25°C (as applicable):

DCL - 200 percent of initial limit.

Cap. - Shall not change more than ±2 percent of initial measured value.

DF - See table I.

ESR - (after 240 and 2,000 hours): See table I.

This test shall be performed on a 100 percent basis following thermal shock and voltage aging of qualification and group A tests.

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At 85°C:

DCL - See table I.

At 125°C:

DCL - See table I.

4,000, 6,000, 8,000, and 10,000 hours:

At 25°C (as applicable):

DCL - 400 percent of initial limit.

Cap. - Shall not change more than ±10 percent of initial measured value.

DF - See table I.

ESR - (after 10,000 hours): See table I.

At 85°C:

DCL - 200 percent of initial limit.

At 125°C:

DCL - 200 percent of initial limit.

Marking: In accordance with MIL-C-39003.

Part number: M39003/09-(dash number from table I).

APPLICATION NOTES:

1. Rated ripple current is the rms value of the maximum allowable alternating current of a specified frequency, at which the capacitor may be operated continuously at a specified temperature. Derate ripple current for ambient temperature in accordance with the curve given on figure 2.

2. For derating the frequency, use the derated ripple currents at 1 kHz given in table I. Below 1 kHz these same currents are applicable provided the peak ac voltage does not result in voltage reversal or exceeding the rated dc voltage. Between 1 kHz and 40 kHz the ripple current may be interpolated linearly with frequency. The ripple current at 40 kHz is applicable at the above 40 kHz.

3. Although CSR21 capacitors are rated to operate with the specified levels of rms ripple current, they are basically polar devices. Care must be exercised to assure that sufficient dc bias is applied to prevent ac voltage reversal in excess of specified reverse voltage ratings (see MIL-STD-198).

4. When two or more CSR21 capacitors are used in parallel, ripple current may not divide equally as a result of unequal ESR's of the capacitors. It is imperative that each capacitor be operated within the specified limit of rms ripple current.

Ⓑ Substitutability data:

Substitutability for failure rate levels shall be as follows: In accordance with MIL-C-39003.

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Ⓑ TABLE I. Electrical characteristics and part numbers.

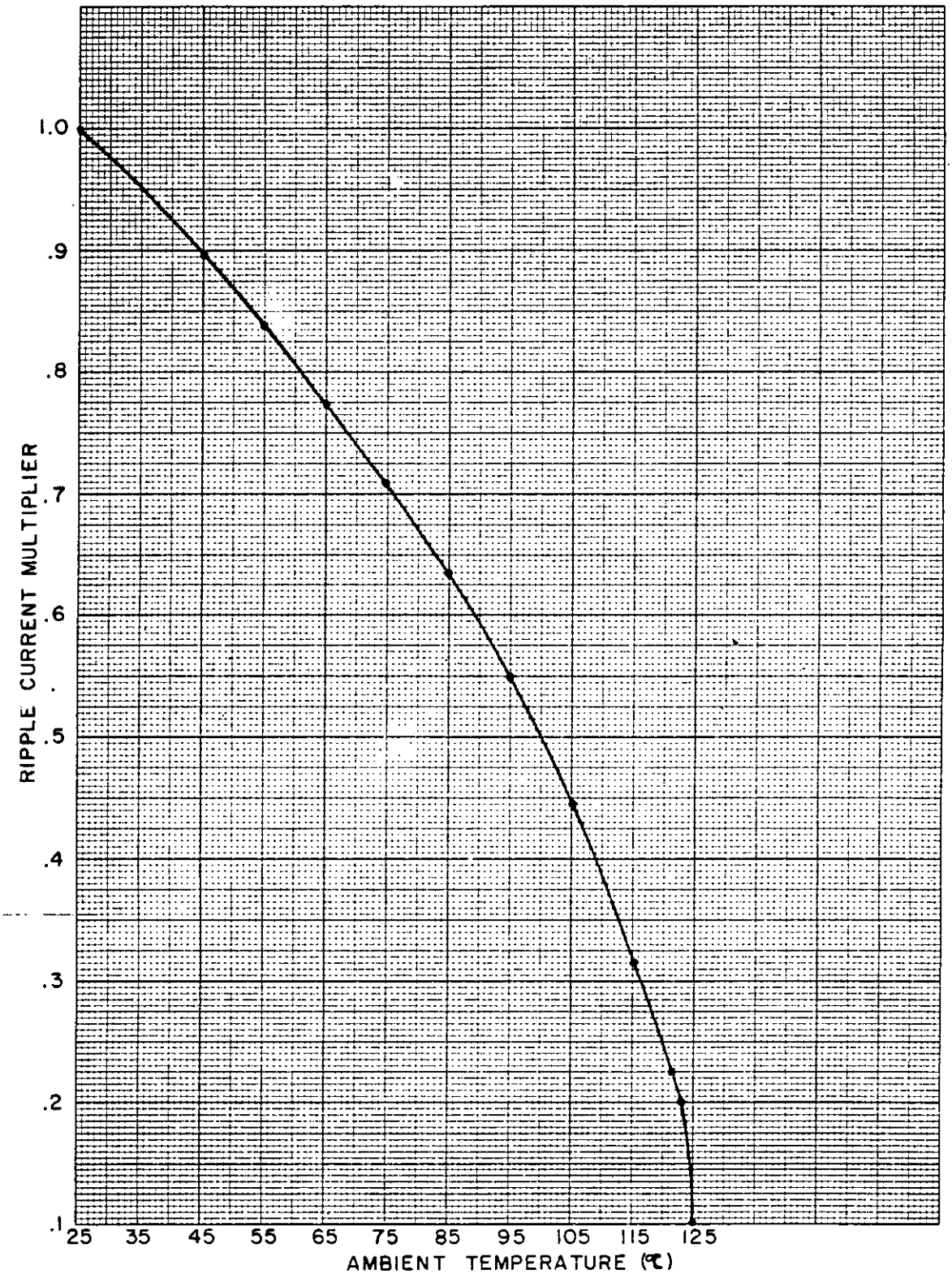
DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at			Dissipation factor 1 kHz +25°C	ESR 100 kHz +25°C (max)	Ripple current 40 kHz 25°C (max)	Derated ripple current 40 kHz 25°C (max)	Case size	Part no. M39003/09- Failure rate level for (%1,000 hrs)						
			+25°C	+85°C	+125°C						M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
Volts	µF	Percent	µA	µA	µA	Percent	Ohms	A									
6	150	5	4.5	90	113	10	.065	3.3	2.0	C	0001	0101	0201	0301	2001	3001	4001
6	150	10	4.5	90	113	10	.065	3.3	2.0	C	0002	0102	0202	0302	2002	3002	4002
6	150	20	4.5	90	113	10	.065	3.3	2.0	C	0003	0103	0203	0303	2003	3003	4003
6	180	5	5.5	110	138	10	.060	3.4	2.4	C	0004	0104	0204	0304	2004	3004	4004
6	180	10	5.5	110	138	10	.060	3.4	2.4	C	0005	0105	0205	0305	2005	3005	4005
6	270	5	6.5	130	163	10	.050	4.1	3.4	D	0006	0106	0206	0306	2006	3006	4006
6	270	10	6.5	130	163	10	.050	4.1	3.4	D	0007	0107	0207	0307	2007	3007	4007
6	330	5	7.5	150	188	12	.045	4.3	3.8	D	0008	0108	0208	0308	2008	3008	4008
6	330	10	7.5	150	188	12	.045	4.3	3.8	D	0009	0109	0209	0309	2009	3009	4009
6	330	20	7.5	150	188	12	.045	4.3	3.8	D	0010	0110	0210	0310	2010	3010	4010
10	82	5	4	80	100	8	.085	2.9	1.8	C	0011	0111	0211	0311	2011	3011	4011
10	82	10	4	80	100	8	.085	2.9	1.8	C	0012	0112	0212	0312	2012	3012	4012
10	100	5	5	100	125	8	.075	3.0	2.2	C	0013	0113	0213	0313	2013	3013	4013
10	100	10	5	100	125	8	.075	3.0	2.2	C	0014	0114	0214	0314	2014	3014	4014
10	100	20	5	100	125	8	.075	3.0	2.2	C	0015	0115	0215	0315	2015	3015	4015
10	120	5	6	120	150	8	.070	3.2	2.5	C	0016	0116	0216	0316	2016	3016	4016
10	120	10	6	120	150	8	.070	3.2	2.5	C	0017	0117	0217	0317	2017	3017	4017
10	180	5	9	180	226	8	.060	3.7	3.4	D	0018	0118	0218	0318	2018	3018	4018
10	180	10	9	180	226	8	.060	3.7	3.4	D	0019	0119	0219	0319	2019	3019	4019
10	220	5	10	200	250	10	.055	3.9	3.4	D	0020	0120	0220	0320	2020	3020	4020
10	220	10	10	200	250	10	.055	3.9	3.4	D	0021	0121	0221	0321	2021	3021	4021
10	220	20	10	200	250	10	.055	3.9	3.4	D	0022	0122	0222	0322	2022	3022	4022
15	56	5	4	80	100	6	.100	2.6	1.8	C	0023	0123	0223	0323	2023	3023	4023
15	56	10	4	80	100	6	.100	2.6	1.8	C	0024	0124	0224	0324	2024	3024	4024
15	68	5	5	100	125	6	.095	2.7	2.2	C	0025	0125	0225	0325	2025	3025	4025
15	68	10	5	100	125	6	.095	2.7	2.2	C	0026	0126	0226	0326	2026	3026	4026
15	68	20	5	100	226	6	.095	2.7	2.2	C	0027	0127	0227	0327	2027	3027	4027
15	120	5	9	180	226	8	.070	3.5	2.8	D	0028	0128	0228	0328	2028	3028	4028
15	120	10	9	180	226	8	.070	3.5	2.8	D	0029	0129	0229	0329	2029	3029	4029
15	150	5	10	200	250	8	.065	3.6	3.1	D	0030	0130	0230	0330	2030	3030	4030
15	150	10	10	200	250	8	.065	3.6	3.1	D	0031	0131	0231	0331	2031	3031	4031
15	150	20	10	200	250	8	.065	3.6	3.1	D	0032	0132	0232	0332	2032	3032	4032
20	27	5	2.5	50	63	5	.145	2.2	1.2	C	0033	0133	0233	0333	2033	3033	4033
20	27	10	2.5	50	63	5	.145	2.2	1.2	C	0034	0134	0234	0334	2034	3034	4034
20	33	5	3.5	70	88	5	.130	2.3	1.4	C	0035	0135	0235	0335	2035	3035	4035
20	33	10	3.5	70	88	5	.130	2.3	1.4	C	0036	0136	0236	0336	2036	3036	4036
20	33	20	3.5	70	88	5	.130	2.3	1.4	C	0037	0137	0237	0337	2037	3037	4037
20	39	5	4.0	80	100	5	.120	2.4	1.7	C	0038	0138	0238	0338	2038	3038	4038
20	39	10	4.0	80	100	5	.120	2.4	1.7	C	0039	0139	0239	0339	2039	3039	4039
20	47	5	4.5	90	113	6	.110	2.5	1.8	C	0040	0140	0240	0340	2040	3040	4040
20	47	10	4.5	90	113	6	.110	2.5	1.8	C	0041	0141	0241	0341	2041	3041	4041

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Ⓑ TABLE I. Electrical characteristics and part numbers - Continued.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at			Dissipation factor 1 kHz +25°C	ESR 100 kHz +25°C (max)	Ripple current 40 kHz 25°C (max)	Derated ripple current 40 kHz 25°C (max)	Case size	Part no. M39003/09- Failure rate level for ($1,000 hrs)						
			+25°C	+85°C	+125°C						M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
Volts	µF	Percent	µA	µA	µA	Percent	Ohms	A									
20	47	20	4.5	90	113	6	.110	2.5	1.8	C	0042	0142	0242	0342	2042	3042	4042
20	56	5	5.5	110	138	6	.100	2.9	2.2	D	0043	0143	0243	0343	2043	3043	4043
20	56	10	5.5	110	138	6	.100	2.9	2.2	D	0044	0144	0244	0344	2044	3044	4044
20	68	5	7	140	175	6	.095	3.0	2.4	D	0045	0145	0245	0345	2045	3045	4045
20	68	10	7	140	175	6	.095	3.0	2.4	D	0046	0146	0246	0346	2046	3046	4046
20	68	20	7	140	175	6	.095	3.0	2.4	D	0047	0147	0247	0347	2047	3047	4047
20	82	5	8	160	200	6	.085	3.1	2.5	D	0048	0148	0248	0348	2048	3048	4048
20	82	10	8	160	200	6	.085	3.1	2.5	D	0049	0149	0249	0349	2049	3049	4049
20	100	5	10	200	250	8	.075	3.3	2.5	D	0050	0150	0250	0350	2050	3050	4050
20	100	10	10	200	250	8	.075	3.3	2.5	D	0051	0151	0251	0351	2051	3051	4051
20	100	20	10	200	250	8	.075	3.3	2.5	D	0052	0152	0252	0352	2052	3052	4052
35	22	5	4	80	100	4	.160	2.1	1.5	C	0053	0153	0253	0353	2053	3053	4053
35	22	10	4	80	100	4	.160	2.1	1.5	C	0054	0154	0254	0354	2054	3054	4054
35	22	20	4	80	100	4	.160	2.1	1.5	C	0055	0155	0255	0355	2055	3055	4055
35	27	5	4.5	90	113	4	.145	2.4	1.9	D	0056	0156	0256	0356	2056	3056	4056
35	27	10	4.5	90	113	4	.145	2.4	1.9	D	0057	0157	0257	0357	2057	3057	4057
35	33	5	5.5	110	138	5	.130	2.5	1.9	D	0058	0158	0258	0358	2058	3058	4058
35	33	10	5.5	110	138	5	.130	2.5	1.9	D	0059	0159	0259	0359	2059	3059	4059
35	33	20	5.5	110	138	5	.130	2.5	1.9	D	0060	0160	0260	0360	2060	3060	4060
35	39	5	7	140	175	5	.120	2.6	2.0	D	0061	0161	0261	0361	2061	3061	4061
35	39	10	7	140	175	5	.120	2.6	2.0	D	0062	0162	0262	0362	2062	3062	4062
35	47	5	8	160	200	5	.110	2.7	2.2	D	0063	0163	0263	0363	2063	3063	4063
35	47	10	8	160	200	5	.110	2.7	2.2	D	0064	0164	0264	0364	2064	3064	4064
35	47	20	8	160	200	5	.110	2.7	2.2	D	0065	0165	0265	0365	2065	3065	4065
50	5.6	5	2.2	45	56	3	.300	1.5	.6	C	0066	0166	0266	0366	2066	3066	4066
50	5.6	10	2.2	45	56	3	.300	1.5	.6	C	0067	0167	0267	0367	2067	3067	4067
50	6.8	5	2.2	45	56	3	.275	1.6	.7	C	0068	0168	0268	0368	2068	3068	4068
50	6.8	10	2.2	45	56	3	.275	1.6	.7	C	0069	0169	0269	0369	2069	3069	4069
50	6.8	20	2.2	45	56	3	.275	1.6	.7	C	0070	0170	0270	0370	2070	3070	4070
50	8.2	5	2.5	50	63	3	.250	1.6	.9	C	0071	0171	0271	0371	2071	3071	4071
50	8.2	10	2.5	50	63	3	.250	1.6	.9	C	0072	0172	0272	0372	2072	3072	4072
50	10	5	2.5	50	63	3	.230	1.7	1.1	C	0073	0173	0273	0373	2073	3073	4073
50	10	10	2.5	50	63	3	.230	1.7	1.1	C	0074	0174	0274	0374	2074	3074	4074
50	10	20	2.5	50	63	3	.230	1.7	1.1	C	0075	0175	0275	0375	2075	3075	4075
50	12	5	3	60	75	3	.210	1.8	1.3	C	0076	0176	0276	0376	2076	3076	4076
50	12	10	3	60	75	3	.210	1.8	1.3	C	0077	0177	0277	0377	2077	3077	4077
50	15	5	4	80	100	3	.190	1.9	1.4	C	0078	0178	0278	0378	2078	3078	4078
50	15	10	4	80	100	3	.190	1.9	1.4	C	0079	0179	0279	0379	2079	3079	4079
50	15	20	4	80	100	3	.190	1.9	1.4	C	0080	0180	0280	0380	2080	3080	4080
50	18	5	4.5	90	113	4	.175	2.0	1.4	C	0081	0181	0281	0381	2081	3081	4081
50	18	10	4.5	90	113	4	.175	2.0	1.4	C	0082	0182	0282	0382	2082	3082	4082
50	22	5	5.5	110	138	4	.160	2.3	1.7	D	0083	0183	0283	0383	2083	3083	4083
50	22	10	5.5	110	138	4	.160	2.3	1.7	D	0084	0184	0284	0384	2084	3084	4084
50	22	20	5.5	110	138	4	.160	2.3	1.7	D	0085	0185	0285	0385	2085	3085	4085

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Ⓐ FIGURE 2. Ripple current derating with respect to temperature.

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Substitutability for capacitance tolerance shall be as follows:

Cap tolerance	Will replace cap tolerance
5%	10%, 20%
10%	20%

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Navy - OS, SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - AS, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-39003/9B	CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), HIGH FREQUENCY, ESTABLISHED RELIABILITY, STYLE CSR21

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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